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|---|---|
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| GitHub Link | https://github.com/venkatasai-eng/MLA0403-Deep-learning/tree/main/Assesments |

# SIMATS ENGINEERING

## CO5 - ASSESSMENT TOOL 3

## Technical Presentation

**COURSE NAME: Deep Learning** **COURSE CODE: MLA04**
**SLOT : B** **Faculty : Dr.Arul Raj A.M**

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**COURSE OUTCOME COVERED**

**CO5:** Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

**Course Outcome Weightage: 35%** **Duration: 90 minutes**
**Assessment Tool Weightage: 50%** **Mark : 25**

**Code of Conduct:**

I Guggilla Venkatasai(192425388)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**  **Name of the student:G.Venkatasai**

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| Criteria | Understanding of Problem Context (2) | Application of Concepts (3) | Problem-Solving Approach **(3)** | Accuracy of Solution **(2)** | Clarity ***(1)*** | **Total (10)** |
|---|---|---|---|---|---|---|
| **Weightage** | ***20 %*** | ***30%*** | ***30%*** | ***20 %*** | ***10 %*** | **100%** |
| **Q1** | | | | | | |
| **Total** | | | | | | |

**Signature of the Faculty:** **Total Marks:**

**PHOTO :**


**QUESTIONS:**

Generative Adversarial Networks Create a technical presentation illustrating the working of Generative Adversarial Networks using the Generator and Discriminator framework. Explain the adversarial training process, loss functions, generation of synthetic data, major challenges, and practical applications. Compare GANs with Autoencoders and justify when GANs are a more suitable choice.

**PRESENTATION ANSWER :**

## Generative Adversarial Networks (GANs)

### 1. Introduction

A Generative Adversarial Network (GAN) is a deep learning model used to generate new synthetic data that resembles real training data. A GAN consists of two neural networks:

- Generator (G): Generates fake/synthetic data from random noise.
- Discriminator (D): Distinguishes between real data and generated data.

The two networks compete with each other during training. The Generator tries to fool the Discriminator, while the Discriminator tries to correctly identify fake data.

**2. Generator–Discriminator Framework**

The basic GAN architecture is:

Random Noise (z)
↓
Generator (G)
↓
Generated/Synthetic Data G(z)
↓
Discriminator (D)
↓
Real or Fake

At the same time:

Real Training Data (x) → Discriminator (D)

The Discriminator receives both real samples and generated samples and produces a probability between 0 and 1:

- 1 → Real
- 0 → Fake

The Generator receives a random vector called latent noise and transforms it into synthetic data.

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**3. Working of GAN**

The working process can be explained in the following stages:

Step 1: Generate Random Noise

A random vector z is sampled from a simple probability distribution such as a Gaussian or uniform distribution.

Step 2: Generator Creates Fake Data

The Generator takes the random noise as input:

G(z) → Synthetic Data

Initially, the generated samples are usually poor in quality.

Step 3: Discriminator Evaluates Data

The Discriminator receives:

- Real samples from the training dataset.
- Fake samples produced by the Generator.

It predicts whether each sample is real or fake.

Step 4: Update the Discriminator

The Discriminator learns to correctly classify real samples as real and generated samples as fake.

Step 5: Update the Generator

The Generator receives feedback through the adversarial loss and changes its parameters so that its generated samples become more difficult for the Discriminator to detect.

Step 6: Repeat Training

The Generator and Discriminator are trained alternately. Over many iterations, the Generator becomes capable of producing increasingly realistic samples.

**4. Adversarial Training**

GAN training is an adversarial game between two networks.

Generator's objective:

Generate samples that the Discriminator classifies as real.

Discriminator's objective:

Correctly distinguish real samples from generated samples.

The original GAN objective is:

$$\min_{G}\max_{D} V(D,G) = E_{x\sim p_{data}}[\log D\,(x)] + E_{z\sim p_z}\left[\log\left(1 - D\big(G(z)\big)\right)\right]$$

Where:

- $G$= Generator
- $D$= Discriminator
- $x$= Real data
- $z$= Random noise
- $G(z)$= Generated data
- $D(x)$= Probability that real data is real
- $D\big(G(z)\big)$= Probability that generated data is real

The training continues until the generated distribution becomes close to the real data distribution.

**5. GAN Loss Functions**

Discriminator Loss

The Discriminator loss is:

$$L_D = -E[\log D\,(x)] - E\left[\log\left(1 - D\big(G(z)\big)\right)\right]$$

The Discriminator tries to:

- Give a high probability to real data.
- Give a low probability to generated data.

Generator Loss

A commonly used non-saturating Generator loss is:

$$L_G = -E\left[\log D\left(G(z)\right)\right]$$

The Generator tries to make:

$$D\big(G(z)\big) \to 1$$

Therefore, the Generator is rewarded when its synthetic samples are classified as real.

**6. Generation of Synthetic Data**

Once GAN training is completed, the Generator can be used independently.

The process is:

Random Noise → Trained Generator → Synthetic Data

For example, a GAN trained on human face images can generate new faces that do not correspond to any specific person in the training dataset.

Different random noise vectors can produce different synthetic outputs.

GANs can generate:

- Images
- Faces
- Objects
- Scenes
- Synthetic training data
- Translated images
- Enhanced images

**7. Major Challenges of GANs**

**1. Mode Collapse**

The Generator may produce only a limited variety of outputs instead of learning the complete data distribution.

**2. Training Instability**

Maintaining the correct balance between Generator and Discriminator is difficult.

**3. Vanishing or Weak Gradients**

If the Discriminator becomes too powerful, the Generator may receive weak learning signals.

**4. Evaluation Difficulty**

It is difficult to evaluate both realism and diversity using a single metric.

**5. Hyperparameter Sensitivity**

GAN performance depends strongly on learning rate, architecture, batch size, normalization and other training parameters.

**6. Computational Cost**

Training high-quality GANs, especially for high-resolution data, can require significant GPU memory and computational resources.

## 8. Practical Applications

GANs have many practical applications:

Image Synthesis

Generation of realistic faces, objects and scenes.

Image-to-Image Translation

Converting images from one domain to another.

Super-Resolution

Generating high-resolution versions of low-resolution images.

Image Inpainting

Filling missing or damaged regions of an image.

Data Augmentation

Generating additional synthetic training samples when real data is limited.

Creative Content Generation

Generating visual content for design, media and other creative applications.

**9. GANs vs Autoencoders**

| Feature | GAN | Autoencoder |
|---|---|---|
| Architecture | Generator + Discriminator | Encoder + Decoder |
| Main objective | Generate realistic synthetic data | Reconstruct input data |
| Training signal | Adversarial feedback | Reconstruction loss |
| Main strength | Realistic and sharp generation | Representation and reconstruction |
| Output | New synthetic samples | Reconstructed samples |
| Typical applications | Image synthesis, translation, enhancement | Compression, reconstruction, anomaly detection |
| Major limitation | Training instability and mode collapse | May produce smoother or less detailed outputs |

An Autoencoder learns to compress an input into a latent representation using an Encoder and then reconstruct the input using a Decoder.

A GAN, in contrast, focuses strongly on making generated samples appear realistic to a Discriminator.

**10. When Are GANs More Suitable?**

GANs are more suitable when the primary requirement is realistic synthetic data generation.

GANs are preferred when:

- High visual realism is important.
- Sharp and detailed images are required.
- New synthetic samples need to be generated.
- Image-to-image translation is required.
- Synthetic data is required for data augmentation.
- The goal is generation rather than reconstruction.

Autoencoders are generally more suitable when the main objective is:

- Data compression
- Reconstruction
- Anomaly detection
- Feature representation
- Learning a compact latent representation
- Therefore:

Realistic Generation → GAN

Reconstruction / Representation → Autoencoder

## 11. Conclusion

Generative Adversarial Networks are powerful generative deep-learning models based on competition between a Generator and a Discriminator. The Generator learns to create synthetic samples, while the Discriminator learns to distinguish real samples from generated ones. Through adversarial training, the Generator progressively produces more realistic data.

GANs are especially useful for image synthesis, image translation, super-resolution, inpainting and synthetic data generation. However, challenges such as mode collapse, training instability, evaluation difficulty and high computational requirements must be considered.

Compared with Autoencoders, GANs are a better choice when high-quality and realistic synthetic generation is the primary objective, while Autoencoders are more suitable for reconstruction, compression and representation learning.

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 Exemplary | Level 3 Proficient | Level 2 Developing | Level 1 Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |